

Regression

Computational Statistics Team Teaching



Simple Regression

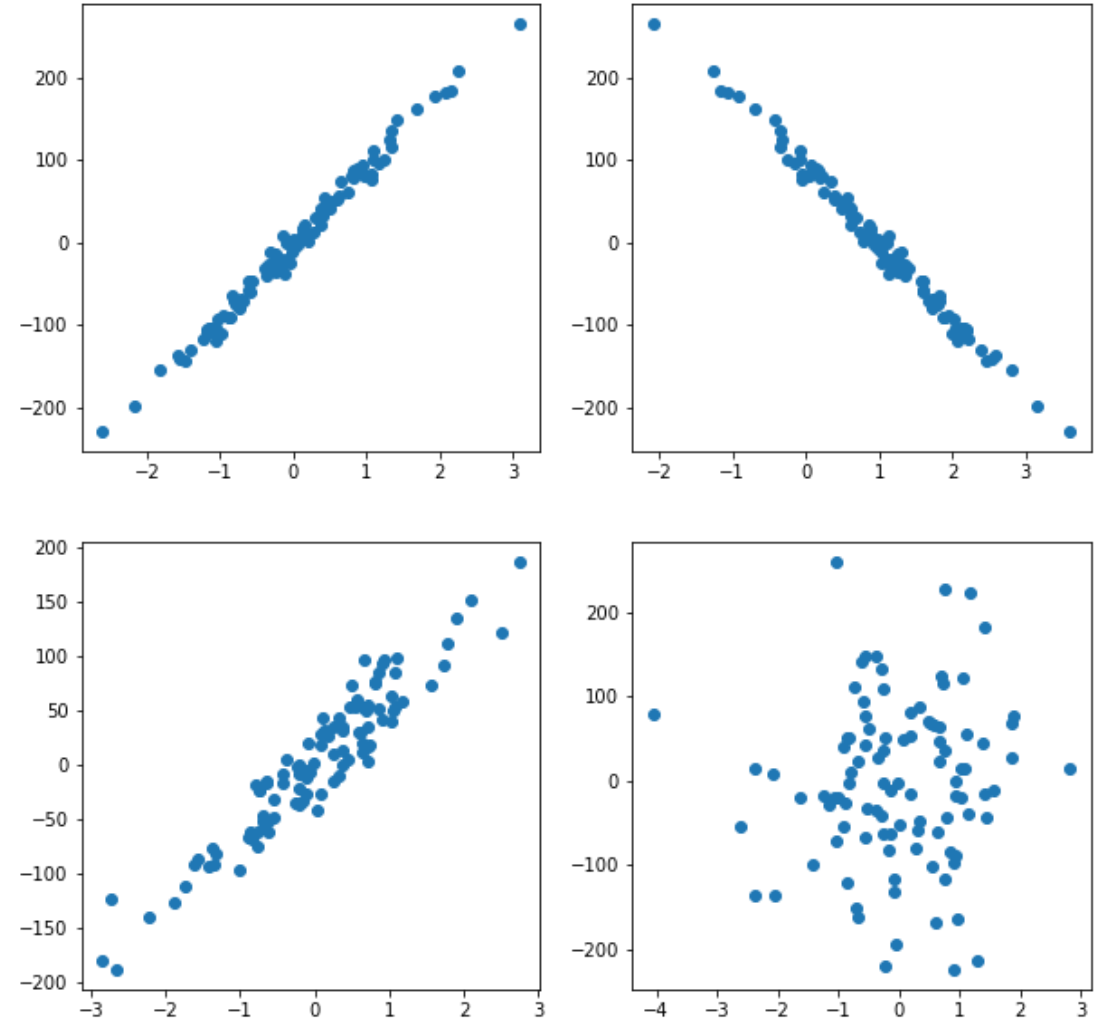
Finding the trend by using single dependent variable

Scatterplot / Scattergraph / Scatter Diagram

A **scatterplot** is a graphical representation of the relationship **between two numerical variables**, where each point represents an observation in the dataset.

Although the data points on a scatterplot may sometimes appear "scattered" or disorganized, we can often visualize an implicit line that suggests a trend within the data.

Approximating Curve
→ Curve Fitting



Regression Line

What is that?

- A line that represents the trend pattern in the data.
- The shape of the line can vary—it may be linear, sigmoid, or parabolic, depending on the data.
- It is also known as the **best-fit line**, **line of best fit**, or **least-squares line**.

IN THIS CLASS WE ONLY FOCUS ON **LINEAR REGRESSION**

Do you remember?



$$y = mx + c$$

Hello straight line equation my old friend! 😊

Simple Linear Regression

What is that?

- A branch of statistics used to analyze the relationship between an **independent variable** and a **dependent variable**.
- In the case of a **linear** relationship, this connection is represented by a straight-line trend.
- It can be utilized to **predict** the value of the dependent variable.

What if

$$\hat{y} = a + bx \quad \text{where}$$

$\hat{y}$ is dependent variable

x is independent variable / exploratory

a is an intercept or y-intercept

b is a *slope* (maybe you call it as gradient)

Simple Linear Regression #1



How to compute a and b ?

$$b = \frac{n \sum xy - \sum x \sum y}{n \sum x^2 - (\sum x)^2}$$

$$a = \frac{\sum y - b \sum x}{n}$$

Example - *b* slope

	x	y
	0	0.8
	2	1.0
	4	0.2
	6	0.2
	8	2.0
	10	0.8
	12	0.6
Sum:	42	5.6

Example - a / y -intercept

	x	y	xy	x^2
	0	0.8	0	0
	2	1.0	2	4
	4	0.2	0.8	16
	6	0.2	1.2	36
	8	2.0	16	64
	10	0.8	8	100
	12	0.6	7.2	144
Sum:	42	5.6	35.2	364

$$a = \frac{\sum y - b \sum x}{n}$$

$$a = \frac{5.6 - \frac{11.2}{784} (42)}{7}$$

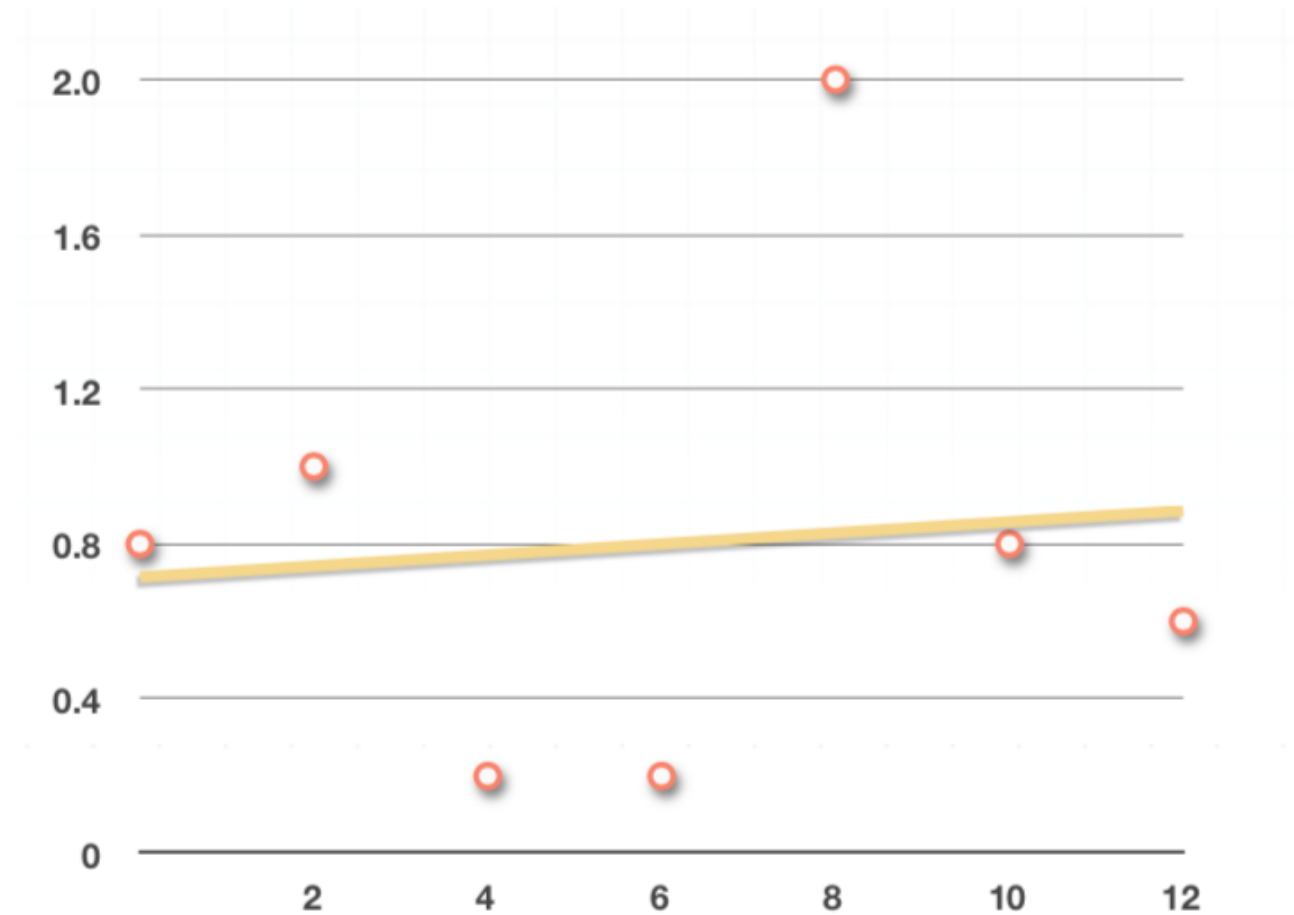
$$a = \frac{5.6 - 0.6}{7} = \frac{5}{7}$$

$$a \approx 0.7143$$

Example – The Regression Line

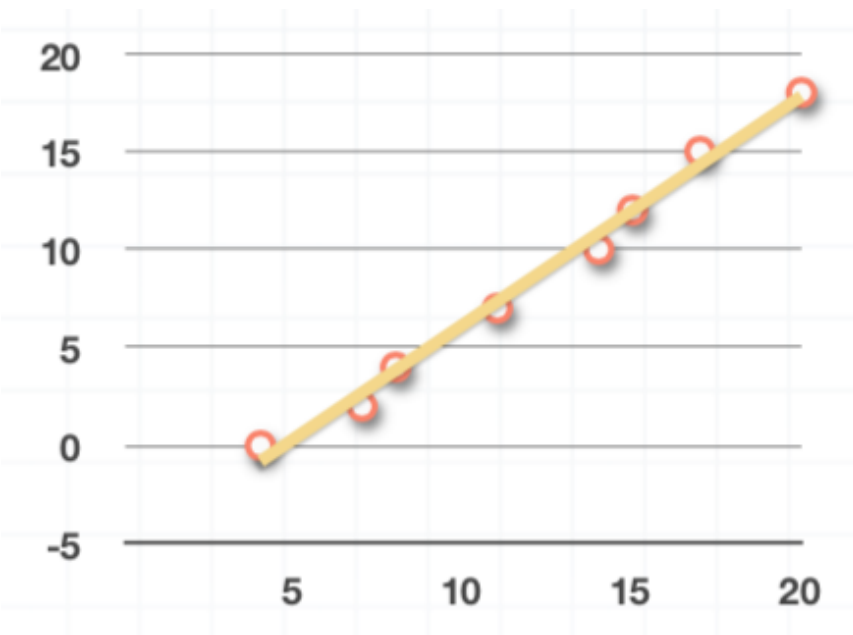
$$\hat{y} = a + bx$$

$$\hat{y} = 0.7143 + 0.0143x$$

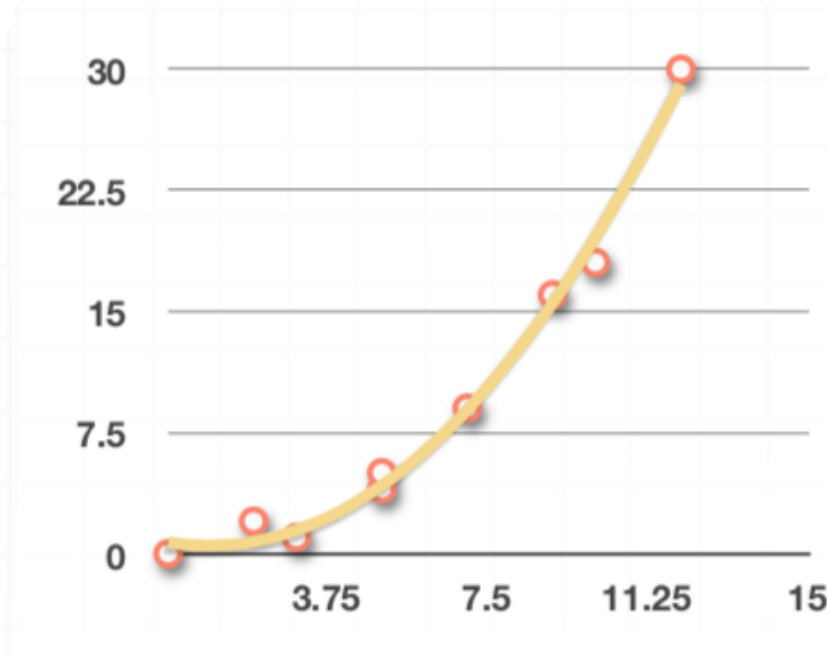


The Trend in Data #1

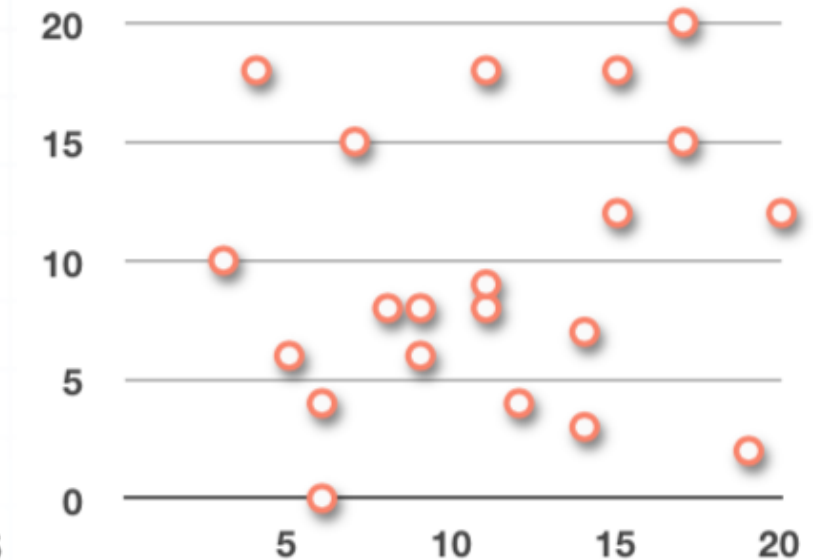
The Trend Shape



Linear



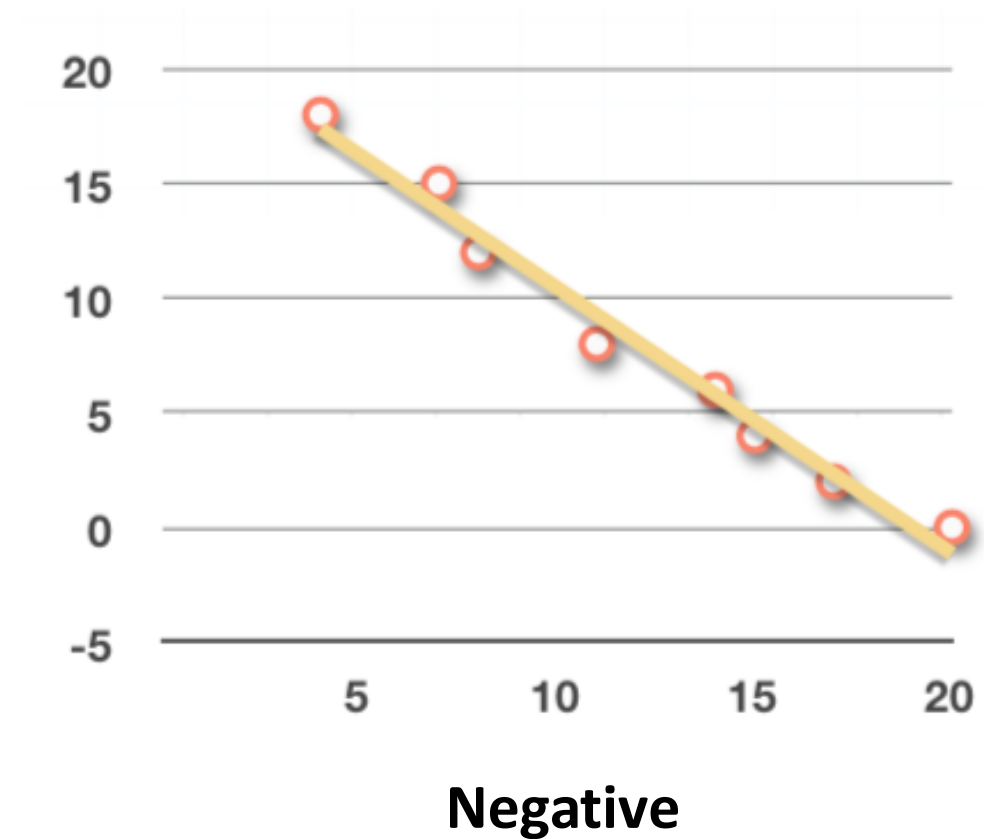
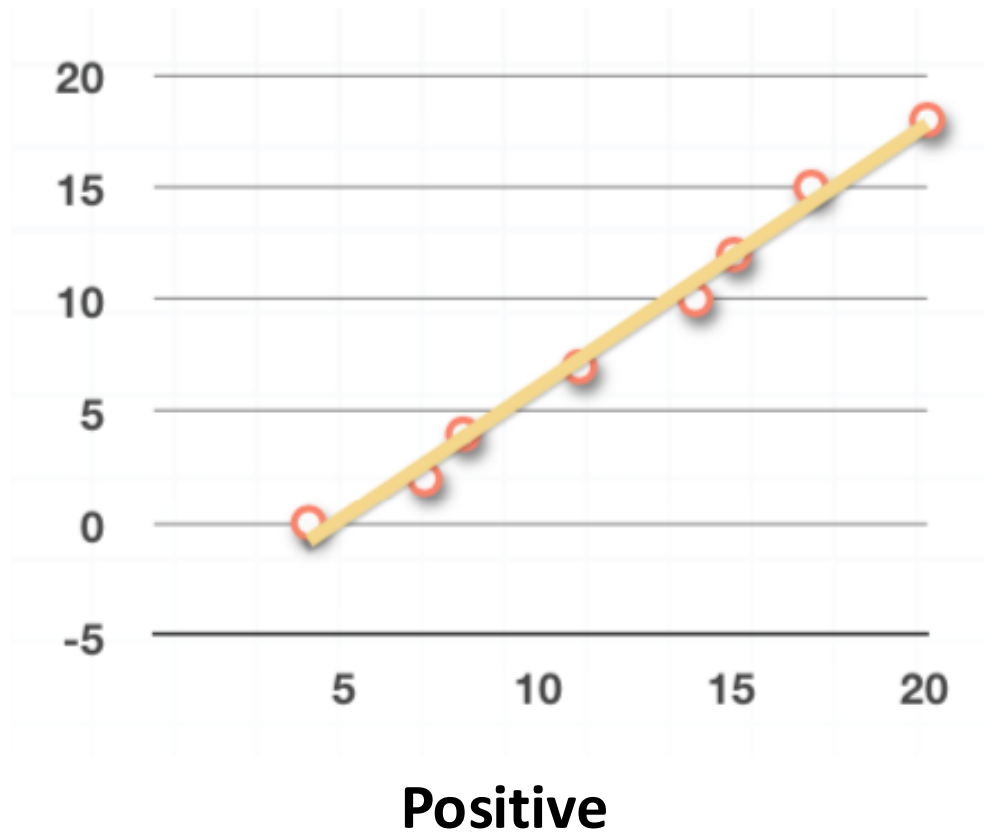
Parabolic



Is it correlated?

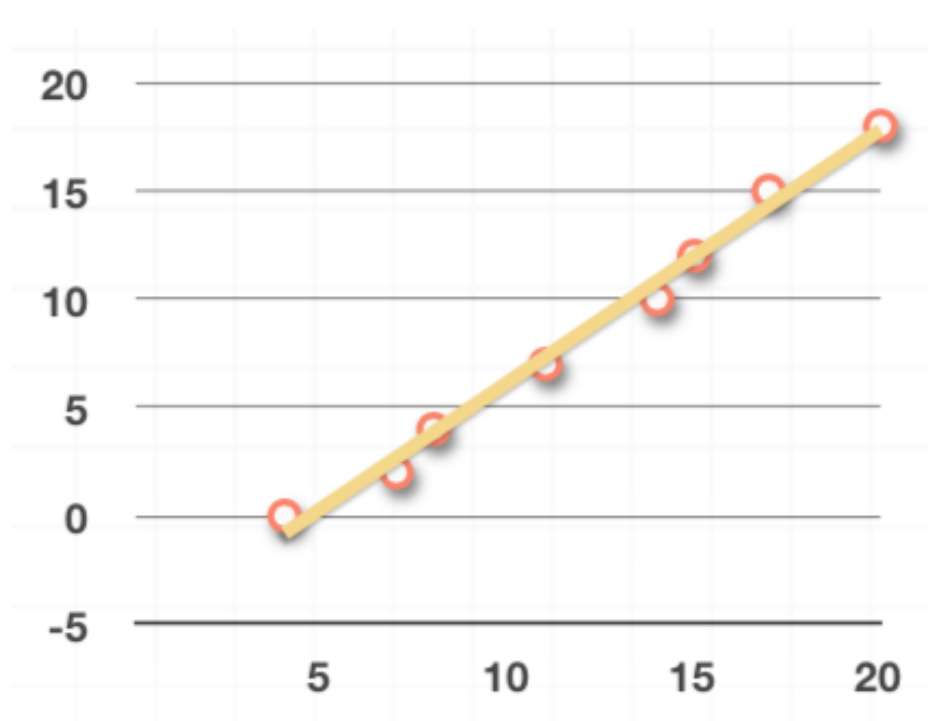
Trend Direction #2

Direction Description

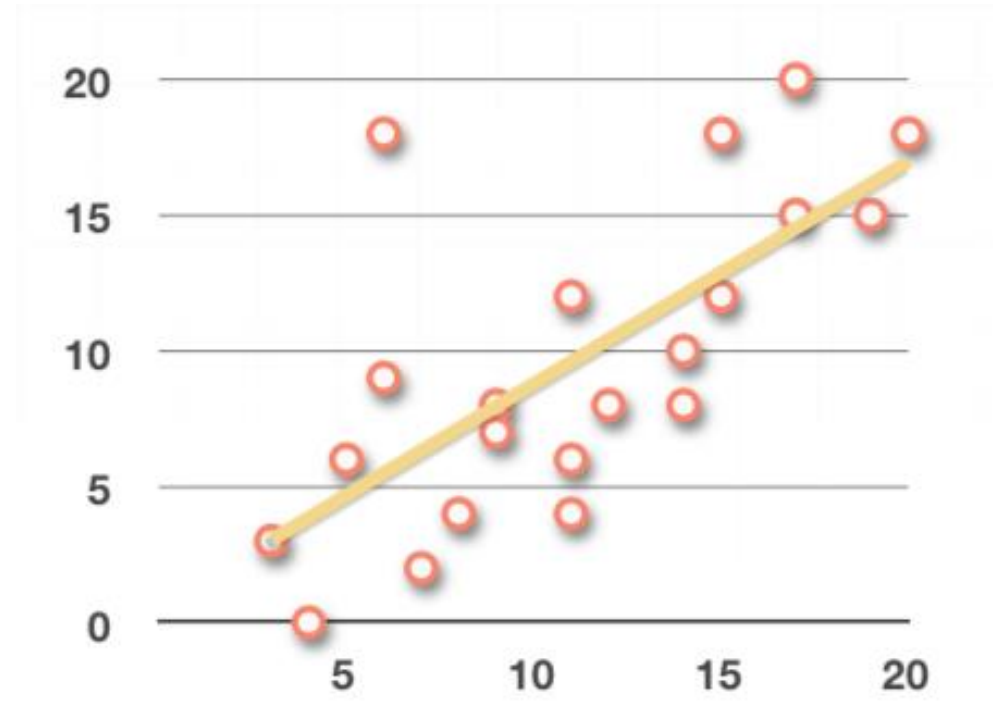


Degree of Correlation

Description of the degree between 2 vars.



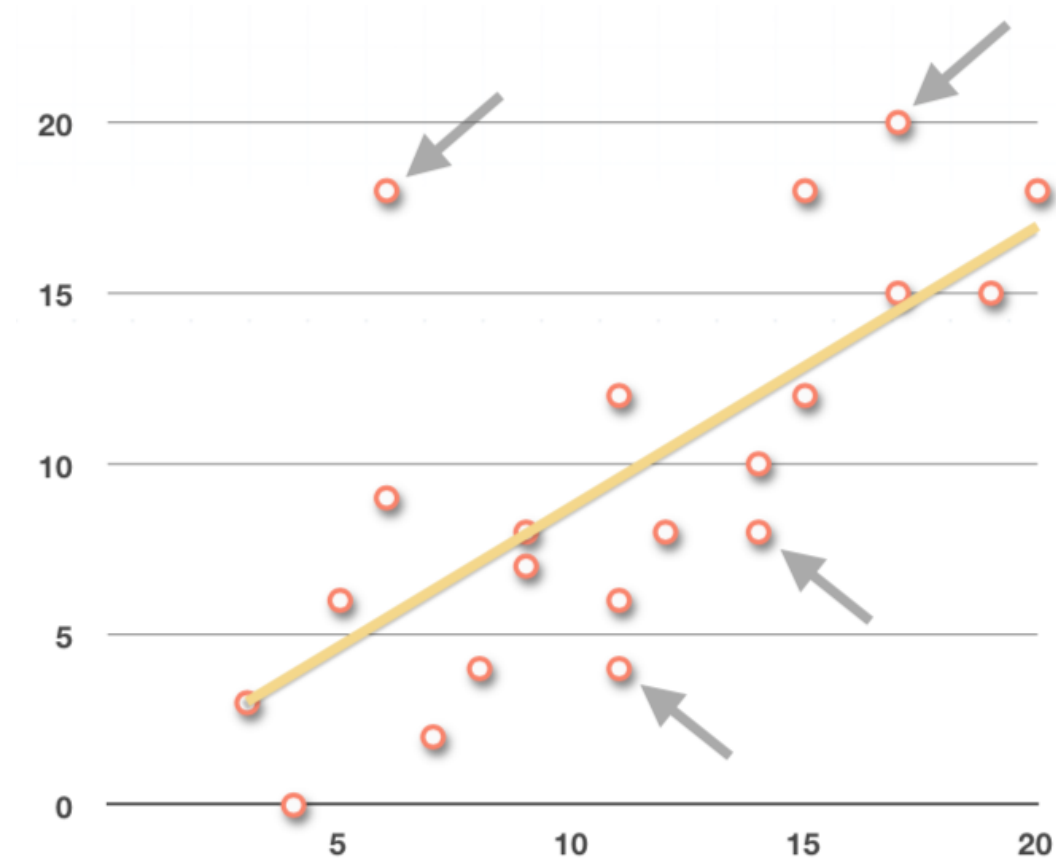
Strong



Weak

Outlier

The outlier of data



Correlation Coefficient

The number to show the degree of correlation between two variables

Correlation Coefficient (r)

Mathematical Desc.

$$r = \frac{1}{n-1} \sum \left(\frac{x_i - \bar{x}}{S_x} \right) \left(\frac{y_i - \bar{y}}{S_y} \right)$$

Diagram illustrating the formula for the Correlation Coefficient (r). The formula is shown with annotations: $\bar{x}$ is labeled "mean x " and S_x is labeled "std x ".

The r can be from 1 to -1
1 shows strong correlation (+)
-1 shows strong correlation (-)

Example

x	y
0	0.8
2	1.0
4	0.2
6	0.2
8	2.0
10	0.8
12	0.6

$$\bar{x} = 6$$

$$\bar{y} = 0.8$$

$$S_x \approx 4.3205$$

$$S_y \approx 0.6110$$

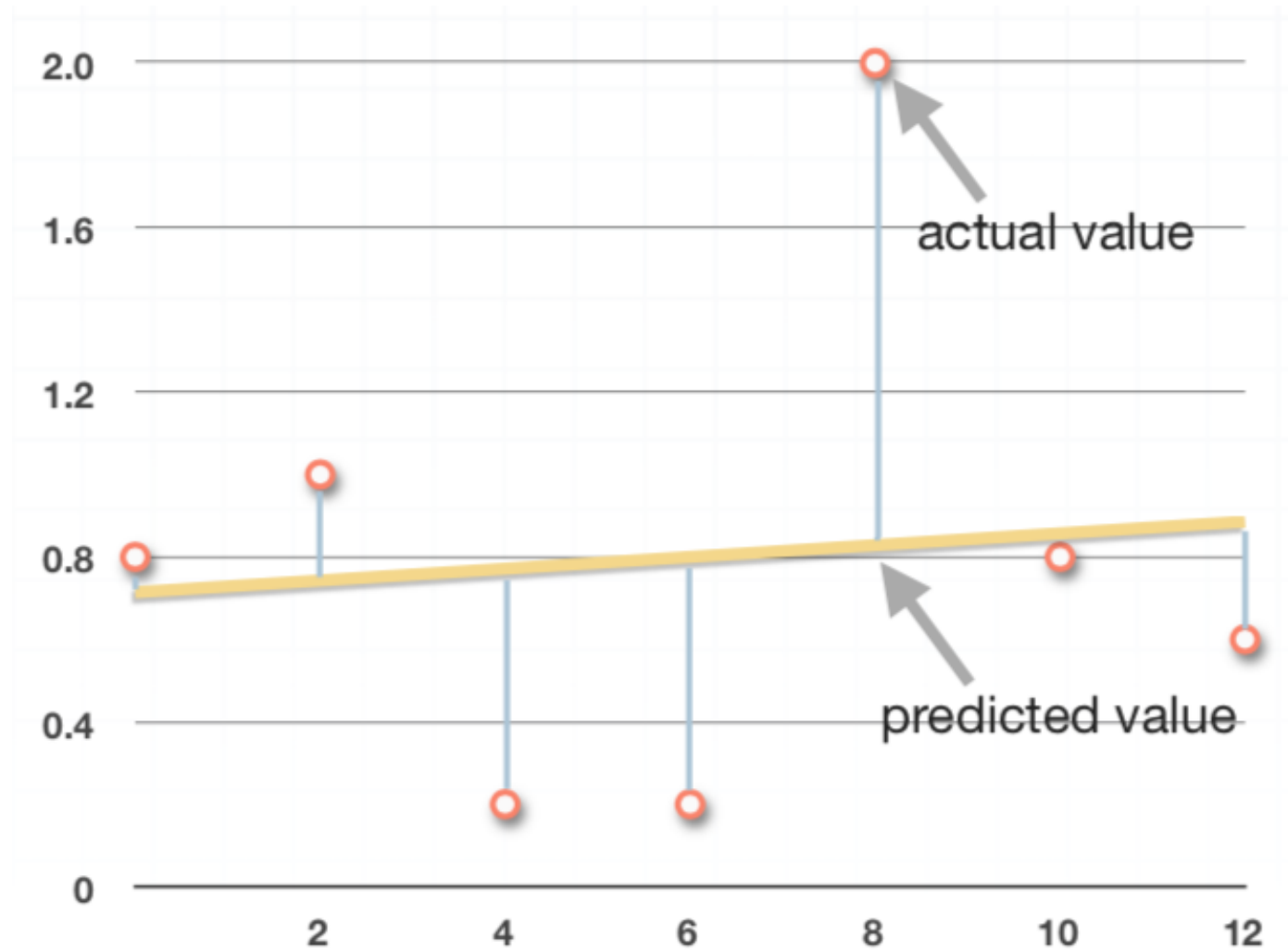
$$r = \frac{1}{7-1} \left(\frac{0-6}{4.3205} \right) \left(\frac{0.8-0.8}{0.6110} \right) + \dots + \left(\frac{12-6}{4.3205} \right) \left(\frac{0.6-0.8}{0.6110} \right)$$

$$r = \frac{1.6}{15.8390} \approx 0.1010$$

Residual (e)

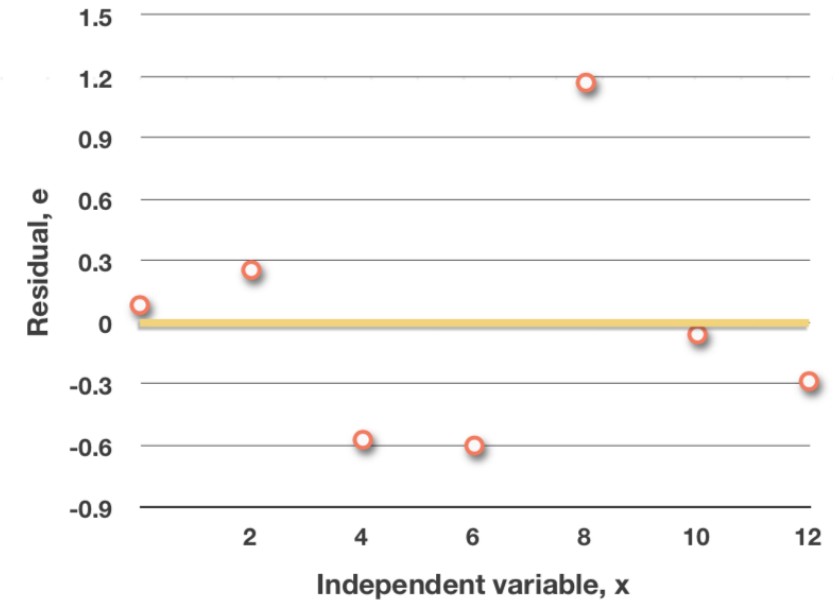
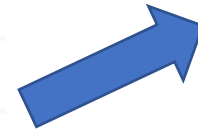
The “gap” actual value and predicted value

Residual (*Error*)



Example

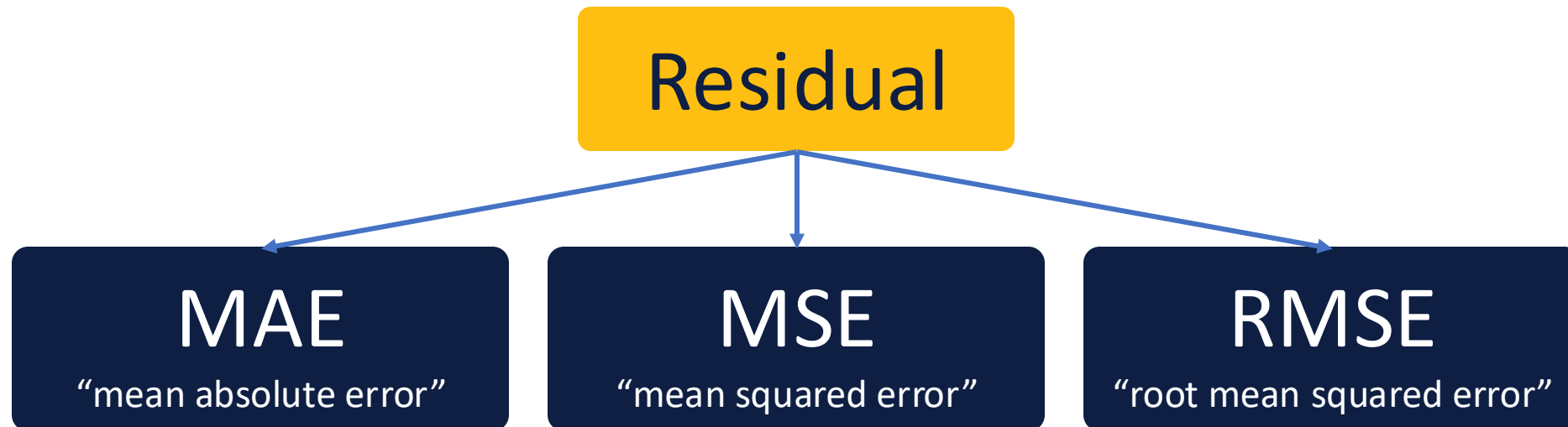
x	Actual	Predicted	e
0	0.8	0.7143	0.0857
2	1.0	0.7429	0.2571
4	0.2	0.7715	-0.5715
6	0.2	0.8001	-0.6001
8	2.0	0.8287	1.1713
10	0.8	0.8573	-0.0573
12	0.6	0.8859	-0.2859



$$\sum e = 0$$

How to define a regression line which fit to the data?

We must identify the regression line model that minimizes the total residual (error). This involves gradually adjusting the **y-intercept** and **slope** until the optimal model is achieved, a process influenced by the **learning rate**.



How to compute residual / error

MAE

“mean absolute error”

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$

MSE

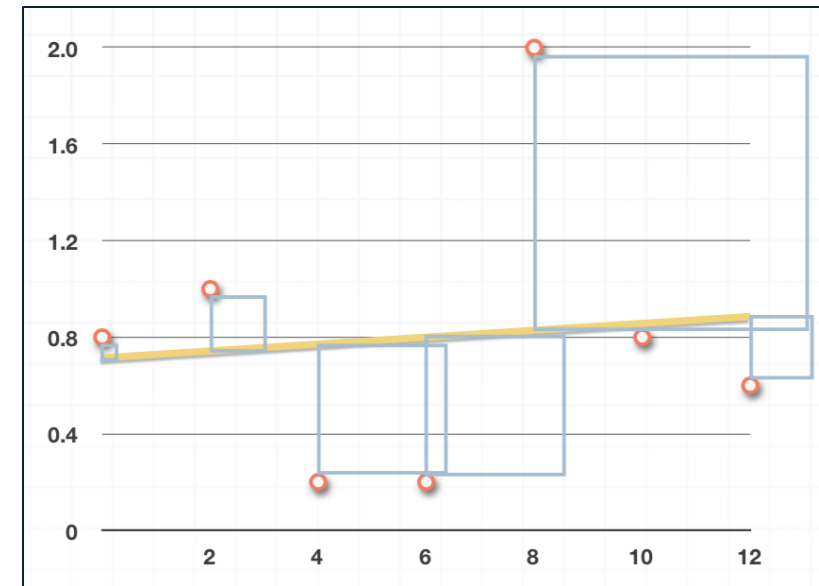
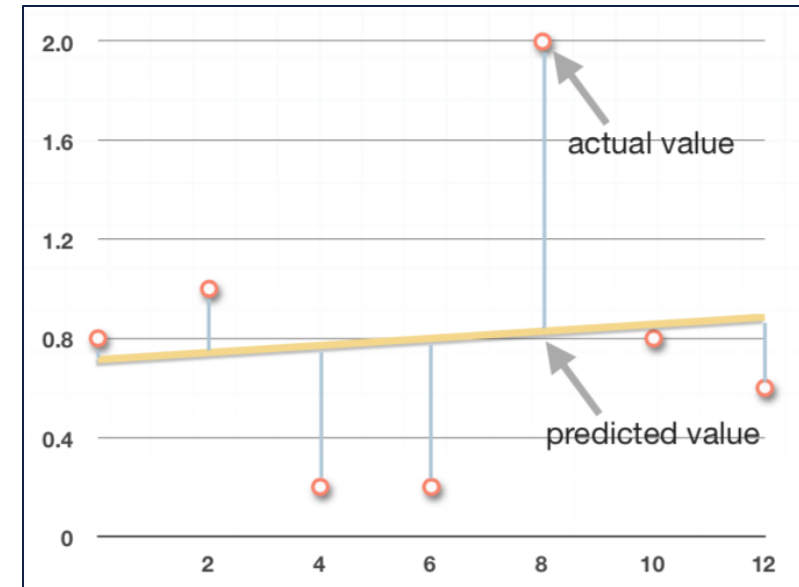
“mean squared error”

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

RMSE

“root mean squared error”

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$$



When we need MAE, MSE, RMSE?

Further reading 😊

- <https://towardsdatascience.com/comparing-robustness-of-mae-mse-and-rmse-6d69da870828>
- <https://medium.com/analytics-vidhya/mae-mse-rmse-coefficient-of-determination-adjusted-r-squared-which-metric-is-better-cd0326a5697e>
- <https://towardsdatascience.com/what-are-rmse-and-mae-e405ce230383>

What if, if we have more than 1 independent variable?

Multiple (Linear) Regression

